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Q1. Short Questions

1) What is Data Set?

1. A **DataSet** is a component of **ADO.NET** used to store data in memory.
2. It works like a **temporary database** inside an application.
3. A DataSet can contain **multiple tables** with rows and columns.
4. It allows applications to work with data **without continuous database connection**.
5. It is mainly used to **store, manage, and manipulate data** before updating the database.

2) What is Data Adapter?

1. A **DataAdapter** acts as a **bridge between the database and the DataSet**.
2. It is used to **retrieve data from the database**.
3. It fills the DataSet using the **Fill() method**.
4. It sends updated data back to the database using the **Update() method**.
5. It works with commands like **Select, Insert, Update, and Delete**.

3) What is the purpose of GET and POST request in Ajax?

1. **GET request** is used to **retrieve data from the server**.
2. The data is **sent through the URL**.
3. It is mainly used when we only **read or fetch information**.
4. GET request is **fast and suitable for small data**.
5. It is **less secure** because the data is visible in the URL.
6. **POST request** is used to **send data to the server**.
7. The data is **sent inside the request body** instead of the URL.
8. It is commonly used for **submitting forms**.
9. POST can **send a large amount of data**.
10. It is **more secure than GET** because the data is not visible in the URL.

4) Write the Difference Between XML and ADO.NET?

XML	ADO.NET
XML is a markup language used to store and transport data.	ADO.NET is a technology used to access and manage databases.
It is platform independent and can be used with different systems.	It is a part of the .NET framework used in .NET applications.
XML stores data in the form of tags and structured format.	ADO.NET is used to connect, retrieve, and manipulate data from databases.
It is mainly used for data sharing between applications.	It is mainly used for database operations in applications.
XML does not directly connect to databases.	ADO.NET provides classes like DataSet and DataAdapter to work with databases.

5) What is the use of WSDL?

1. **WSDL (Web Services Description Language)** is an **XML-based language** used to describe web services.
2. It tells **what services are available** on a web server.
3. It explains **what functions or operations the web service can perform.**
4. It defines **how the client can communicate with the web service.**
5. It also specifies the **message format and communication protocol** used by the web service.

6) What are jQuery Selectors? Define .class selector.

1. **jQuery selectors** are used to **select HTML elements** on a web page.
2. They help developers **manipulate elements** like hide, show, or change content.
3. jQuery selectors work similar to **CSS selectors.**
4. Common selectors are **#id, .class, and element selectors.**
5. The **.class selector** selects all elements that have a specific class name.

Example:

```
$(".menu").hide();
```

Q2. Long Questions

(a) Write the basic syntax of JavaScript. Also write a JavaScript program to convert temperature from Fahrenheit to Celsius.

Answer

Introduction to JavaScript

JavaScript is a **high-level programming language** used to make web pages **interactive and dynamic**. It is mainly used in web development to perform tasks like **form validation, calculations, animations, and dynamic content updates**. JavaScript code is usually written inside the **<script> tag** in an HTML document. It runs in the **web browser** and executes instructions step by step.

Basic Syntax of JavaScript

The basic syntax of JavaScript includes **variables, statements, functions, and output methods**. Variables are used to store data values, and statements perform actions. JavaScript is **case-sensitive**, meaning that uppercase and lowercase letters are treated differently.

Example of basic syntax:

```
<script>
```

```
let a = 10;
```

```
let b = 20;
```

```
let sum = a + b;
```

```
document.write("Sum is: " + sum);
```

```
</script>
```

Explanation:

- `let` is used to declare variables.
 - Variables **a** and **b** store values.
 - `sum` stores the result of addition.
 - `document.write()` is used to display output on the web page.
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JavaScript Program to Convert Fahrenheit to Celsius

Temperature can be converted from Fahrenheit to Celsius using the formula:

$$C = (F - 32) \times 5/9$$

Example program:

```
<script>
```

```
function convertTemp(fahrenheit)
```

```
{
```

```
    let celsius = (fahrenheit - 32) * 5/9;
```

```
    return celsius;
```

```
}
```

```
let f = 100;
```

```
let result = convertTemp(f);
```

```
document.write("Temperature in Fahrenheit: " + f + "<br>");
```

```
document.write("Temperature in Celsius: " + result);
```

```
</script>
```

Explanation:

- A function **convertTemp()** is created to convert temperature.
 - The formula is applied to calculate Celsius.
 - The result is stored in a variable and displayed on the web page.
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Conclusion

JavaScript is an important language used for creating **interactive web applications**. With simple syntax and built-in functions, it can perform calculations and update web page content dynamically.

(b) What is SOAP? Explain SOAP Envelope, Header, Body, and Fault.

Answer

Introduction to SOAP

SOAP (**Simple Object Access Protocol**) is a protocol used for **exchanging information between applications over the internet**. It is based on **XML format** and is commonly used in **web services**. SOAP allows different systems and programming languages to communicate with each other easily. It works with protocols such as **HTTP and SMTP**.

A SOAP message has four main parts: **Envelope, Header, Body, and Fault**.

1) SOAP Envelope

The **SOAP Envelope** is the outermost element of a SOAP message. It defines the **start and end of the message** and indicates that the message is in SOAP format. All other elements such as header and body are contained inside the envelope.

Example:

<Envelope>

</Envelope>

2) SOAP Header

The **SOAP Header** contains additional information related to the message. It may include **security information, authentication details, or transaction data**. The header is optional and is used when extra processing is required.

Example:

<Header>

Authentication Information

</Header>

3) SOAP Body

The **SOAP Body** contains the **actual data or request message** sent between the client and the server. It includes the operation that the web service needs to perform.

Example:

<Body>

Request Data

</Body>

4) SOAP Fault

The **SOAP Fault** element is used to provide **error information** if something goes wrong while processing the SOAP message. It helps identify the problem and gives details about the error.

Example:

<Fault>

Error Message

</Fault>

Conclusion

SOAP is widely used in **web services for communication between applications**. Its structured XML format makes it reliable for sending and receiving data across different systems